

Starlink: A Technical Overview of the SpaceX LEO Broadband System

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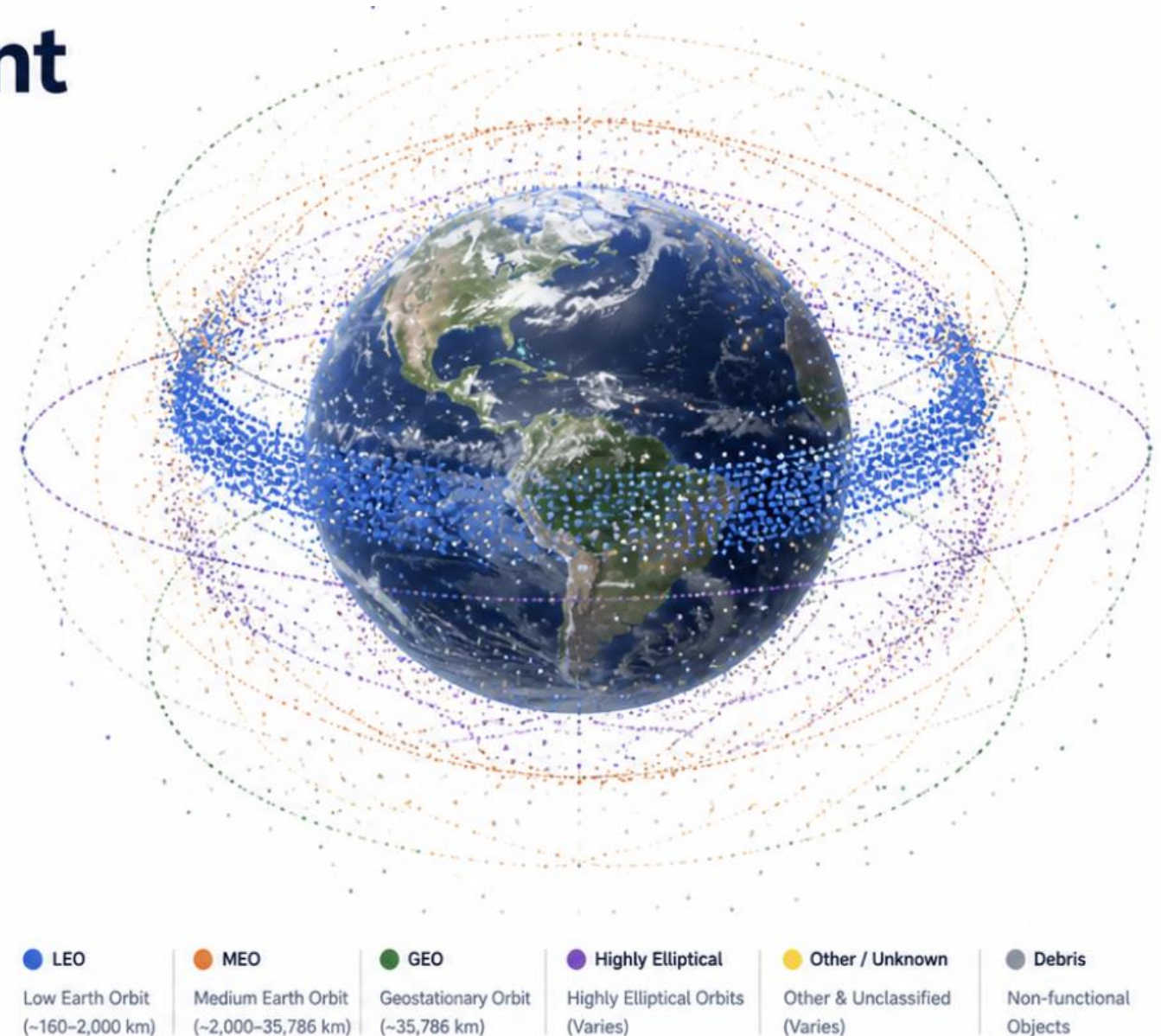
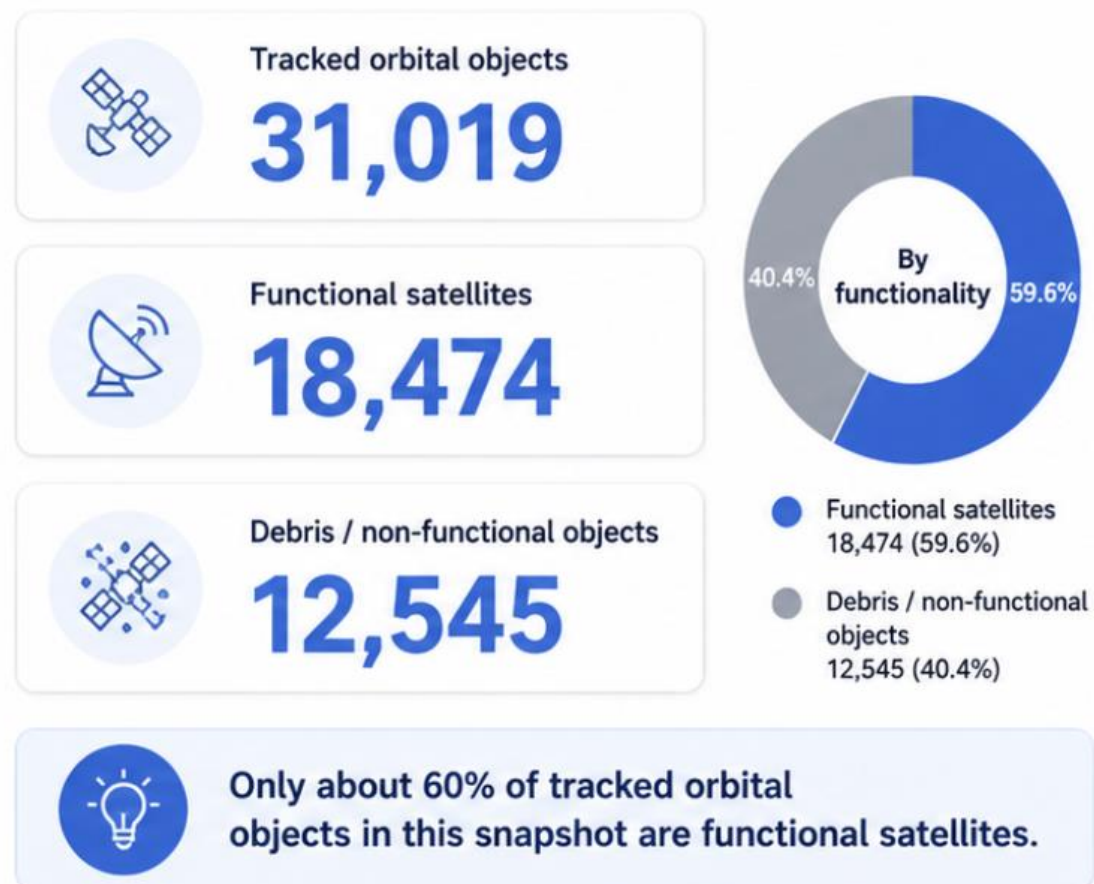
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Overview

The Orbital Environment at a Glance

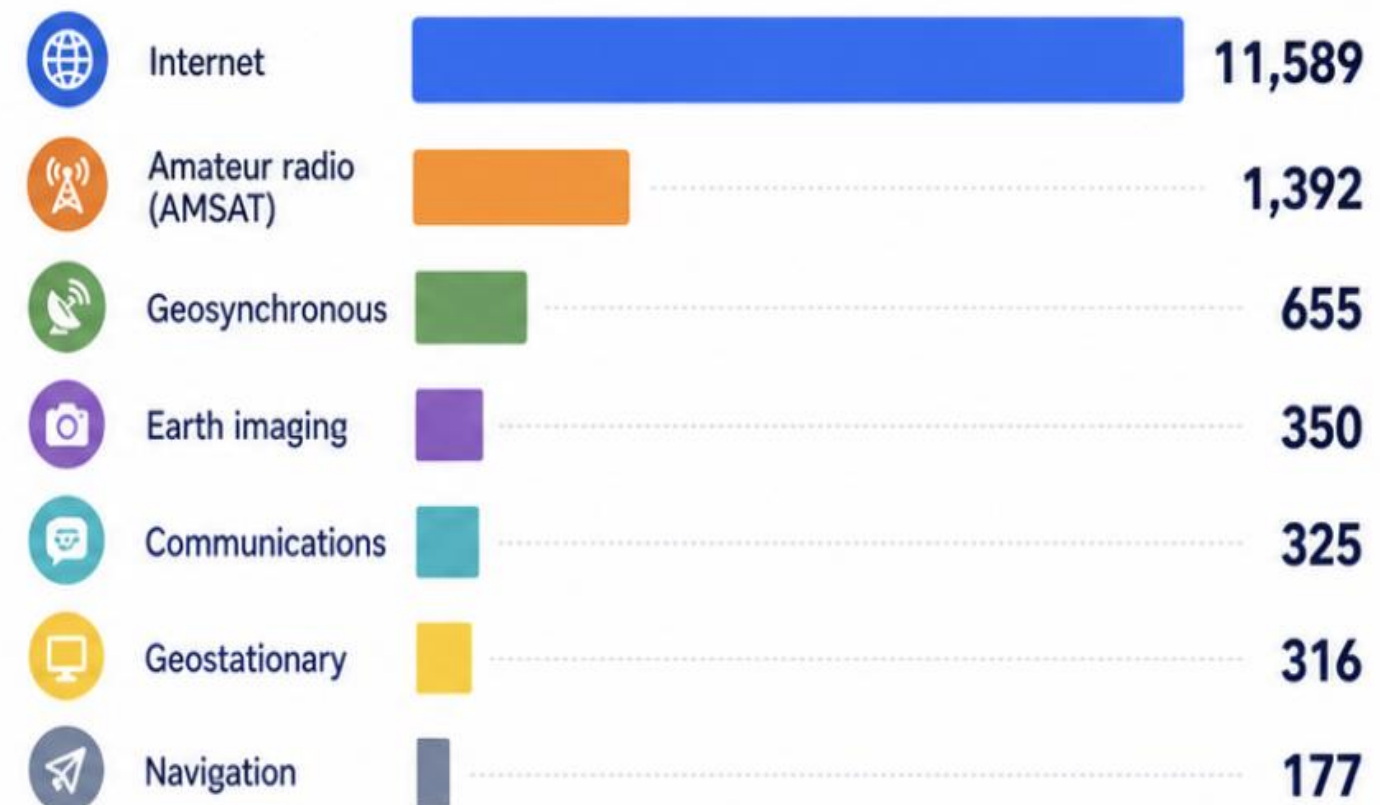
A quick snapshot before we zoom into Starlink



Tasks of Different Satellites

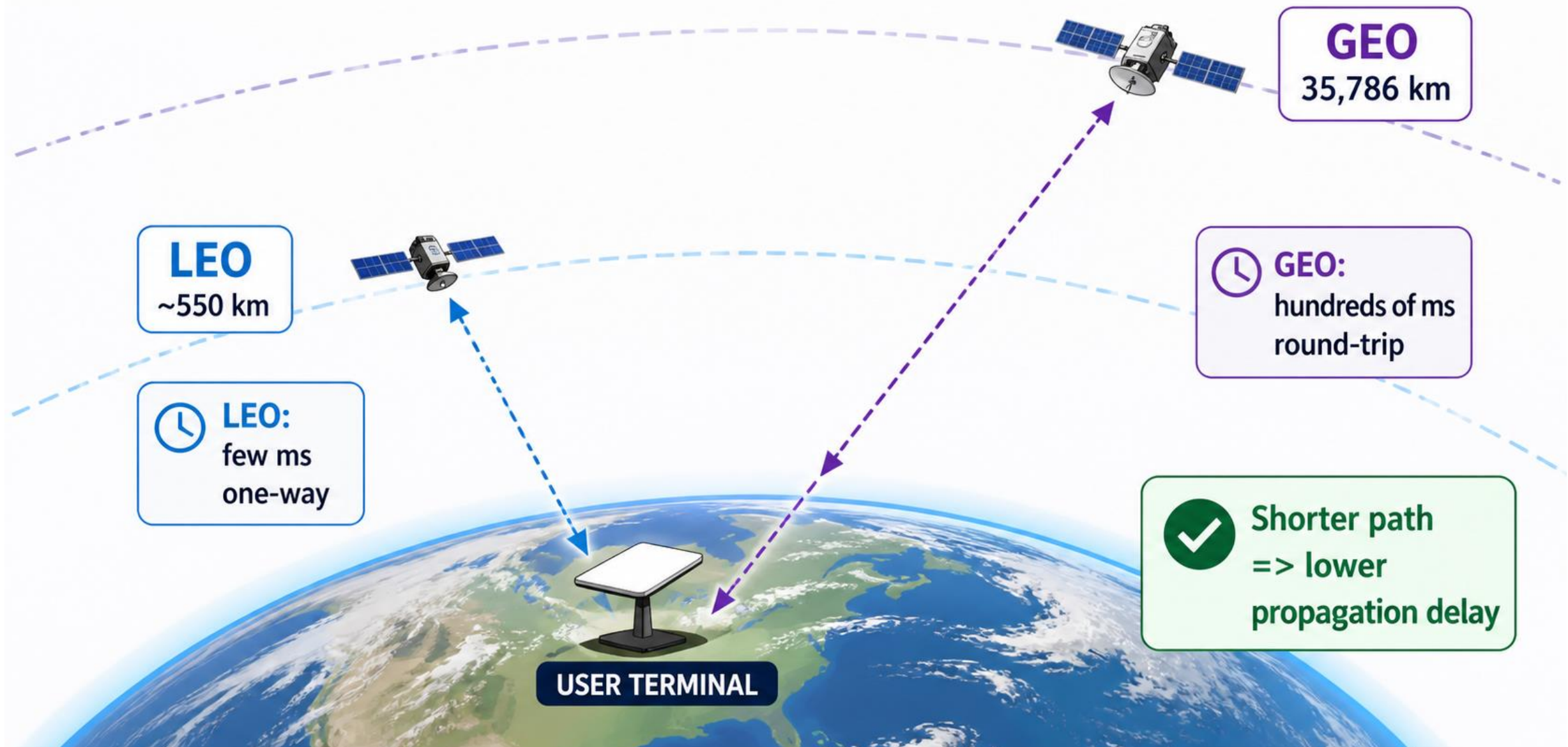


Functional-satellite counts by category

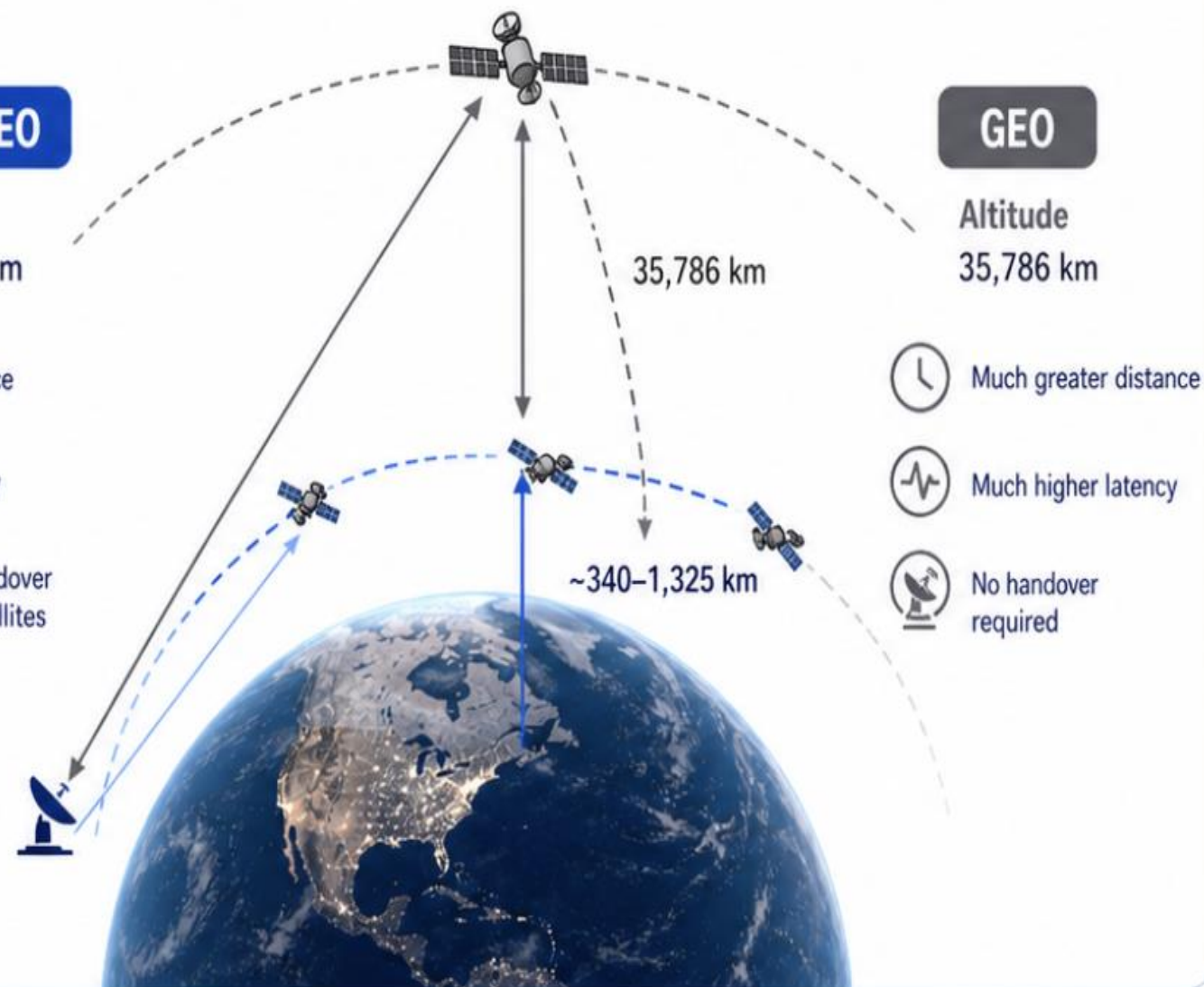


LEO Satellite Background (1/2)

Why LEO Is Different: LEO vs GEO



LEO Satellite Background (2/2)



Representative orbital facts



Altitude range: ~340–1,325 km

Representative public shell examples / authorizations



Inclination range: ~53°–97.6°

Multi-shell design



Orbital speed: ~7.3–7.7 km/s



Orbital period: ~91–107 min



Representative shell: 550 km, 53°

72 planes × 22 satellites per plane



One-way propagation at 550 km: ~1.83–3.74 ms

Zenith to 25° elevation



GEO reference: 35,786 km

Minimum RTT ~477 ms; practical GEO service often >600 ms

Roadmap

Starlink LEO Satellite

Overview of all Satellites and their Purpose

LEO Satellite Background

How Starlink Works and its Features

Subsystems

Interesting Components and Future Works

Why Starlink? (1/2)

A LEO broadband system designed to connect users that are hard to reach with terrestrial infrastructure.



Primary use

Broadband Internet access for rural, remote, and underserved regions.



Why it matters

In the U.S., about 82 million people live in rural areas, and roughly 22 million are offline or underserved.



Why LEO helps

Lower orbital altitude reduces propagation delay compared with geostationary systems.



Core architecture

User terminal → satellite → gateway / PoP → Internet.



Key advantage: Starlink combines wide-area coverage with lower latency than traditional GEO satellite Internet.

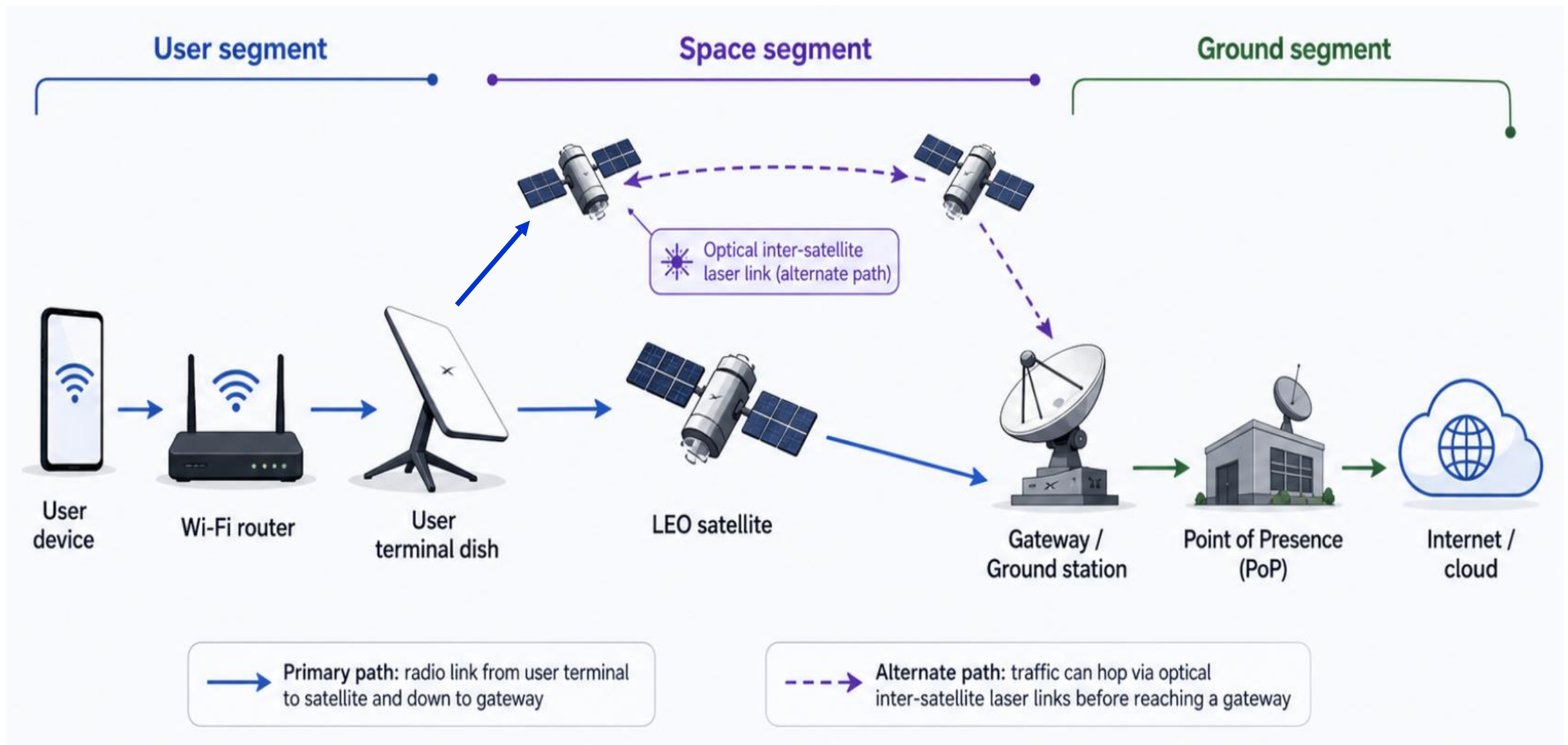
Why Starlink? (2/2)



Starlink by Numbers



How Starlink Works

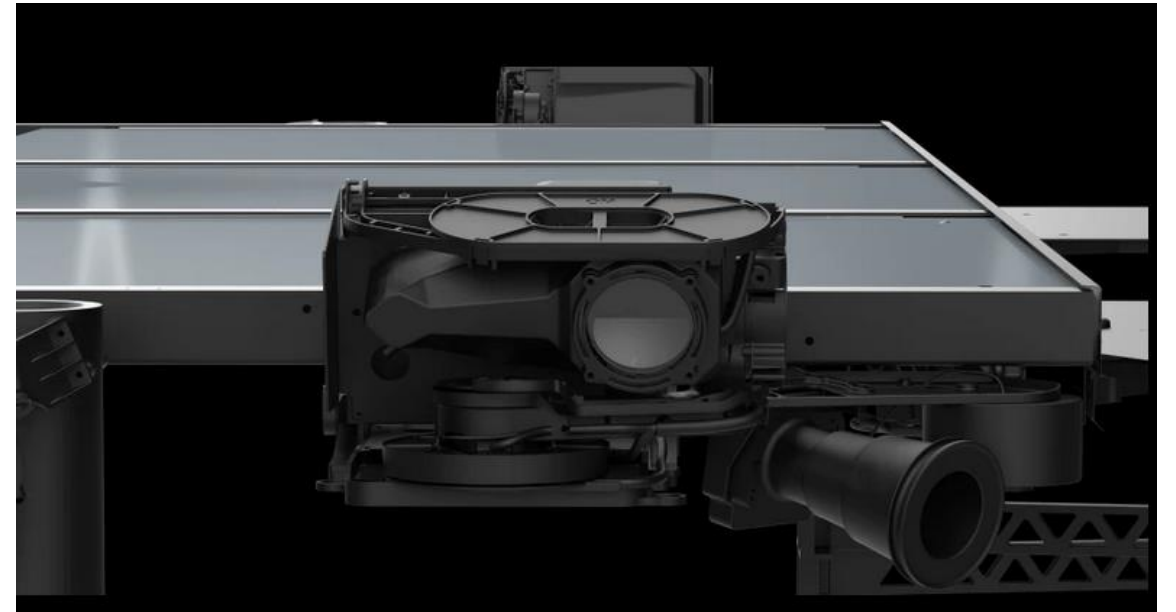


Satellite Subsystems

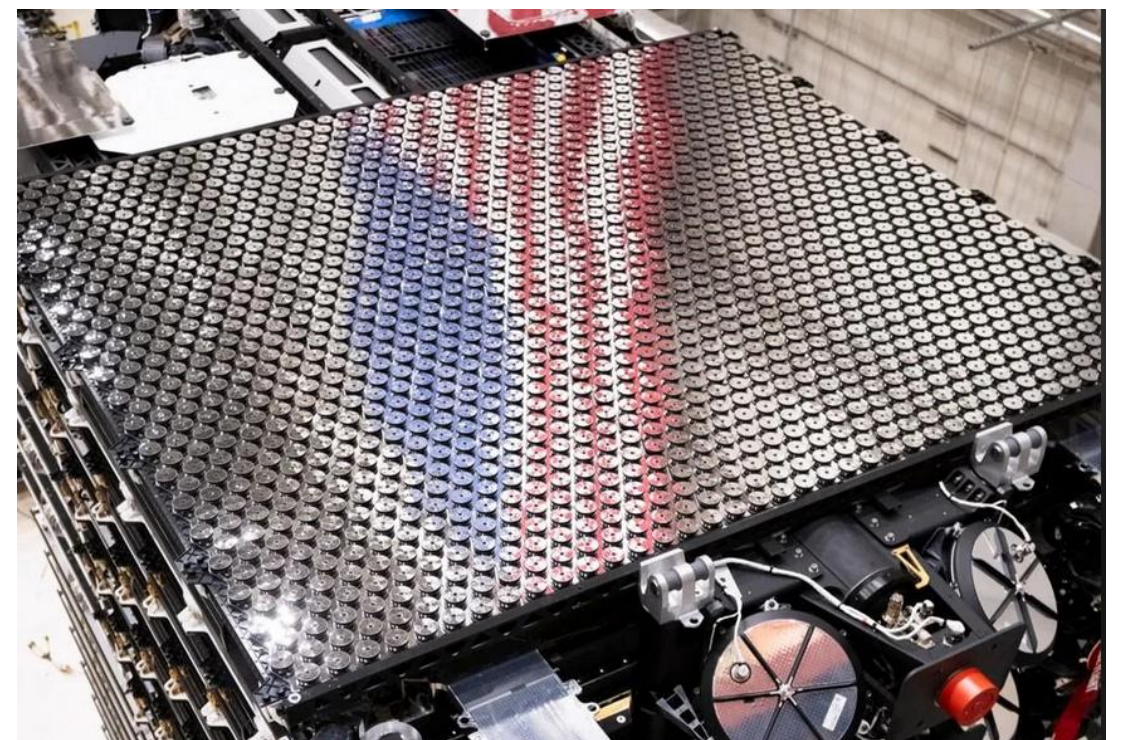
- Communications
- Power
- Attitude and Orbit Control System (AOCS) and Propulsion
- Telemetry, tracking, and command (TT&C)

Communications

- Ku-band phased array antennas
- Dual-band Ka / E-band antennas
 - E-band used for backhauling
- 3 Optical Intersatellite Links (ISL) lasers



Optical Intersatellite Link



Phased Array Antenna

Power Subsystem

- 2 deployable solar arrays
- High-capacity batteries
 - 95.5 minute orbital period, repeatedly enters eclipse
- Batteries support payload and control subsystem operation



Starlink V2 mini dual solar arrays

AOCS and Propulsion

- Argon Hall-effect thrusters
 - 170 mN thrust
 - Used for orbit raising, station-keeping, collision avoidance, end-of-life deorbiting
- 4 Reaction wheels
 - Attitude control via spin



Argon Thruster Ion Propulsion System



Reaction Wheels

TT&C Subsystem

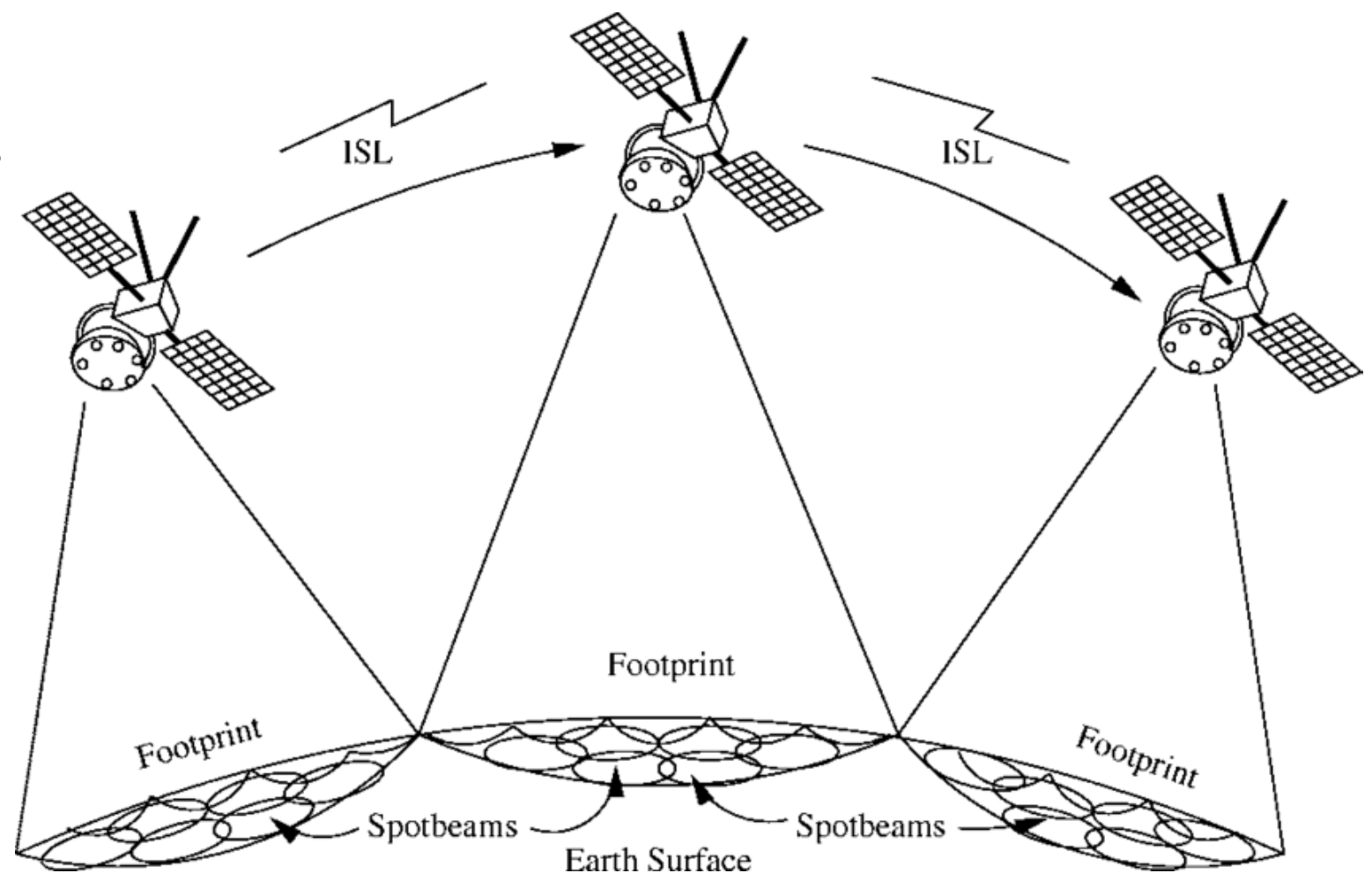
- Utilize VHF bands
 - 137-138 MHz space-to-Earth
 - 148-150.5 MHz Earth-to-space
- Limited to 450 Gen2 satellites at one time
- Star Tracker navigation sensors
 - Provide location, altitude, and orientation
 - Used for determining precise array tuning for broadband throughput



Star Tracker

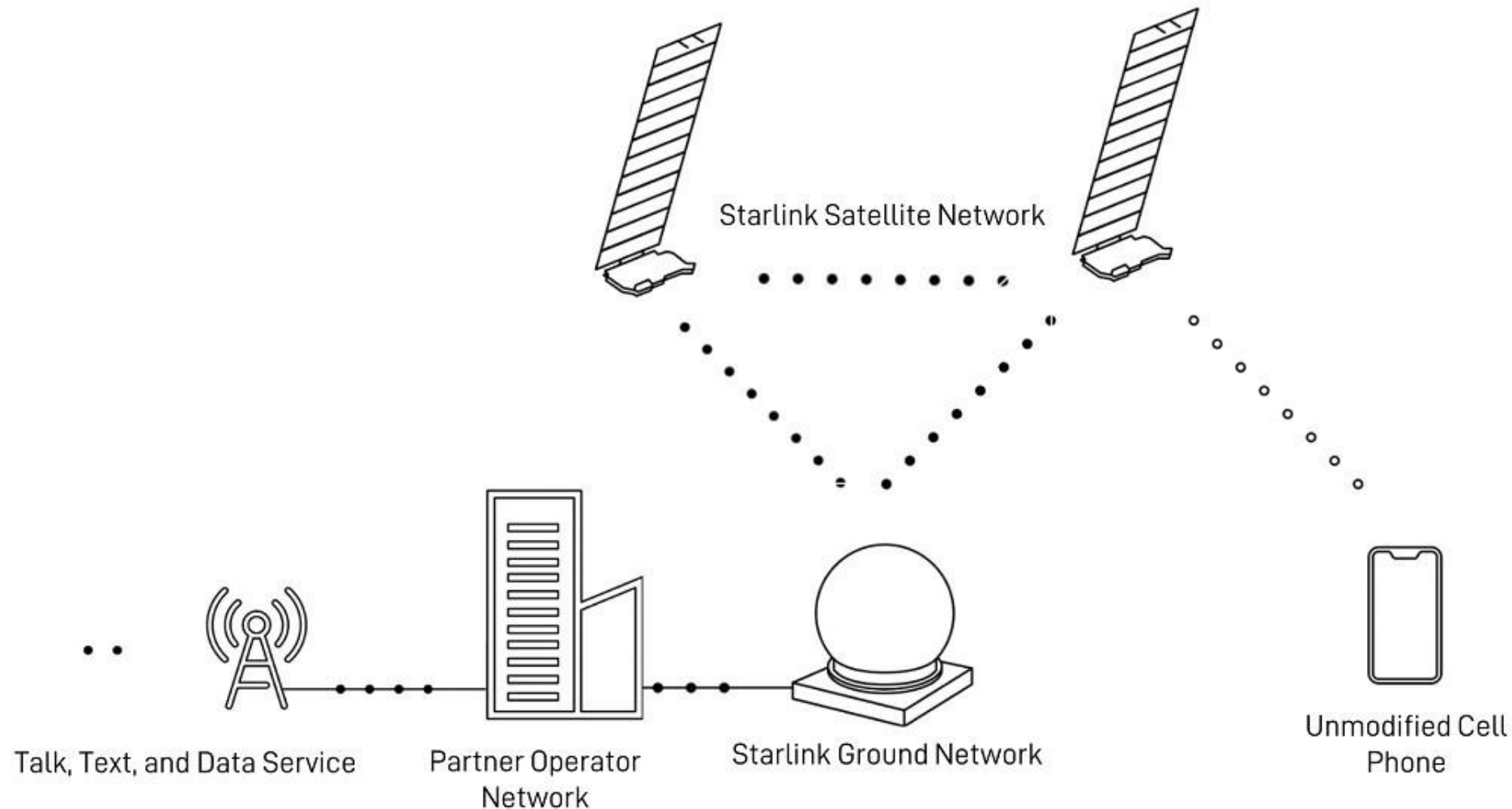
Interesting Components

- Beam Steering:
 - Phased Array Antenna makes ≈ 48 steerable spot beams.
 - Allows Frequency Reuse
 - High aggregate Throughput
 - Capacity directed to high-demanding areas.
- Narrow Bandwidth (2.5°)
- Antennas handover speeds.



Future Work

- Higher-capacity satellite generations, with V2 and V3.
- Direct-to-Cell Expansion.



Future Work

- Commercial Scaling and Sustainability
 - 200 Starlink launches, 430 Falcon 9 Rocket launches
 - Projected in 2026, \$20 Billion Revenue, 17 Million Subscribers.

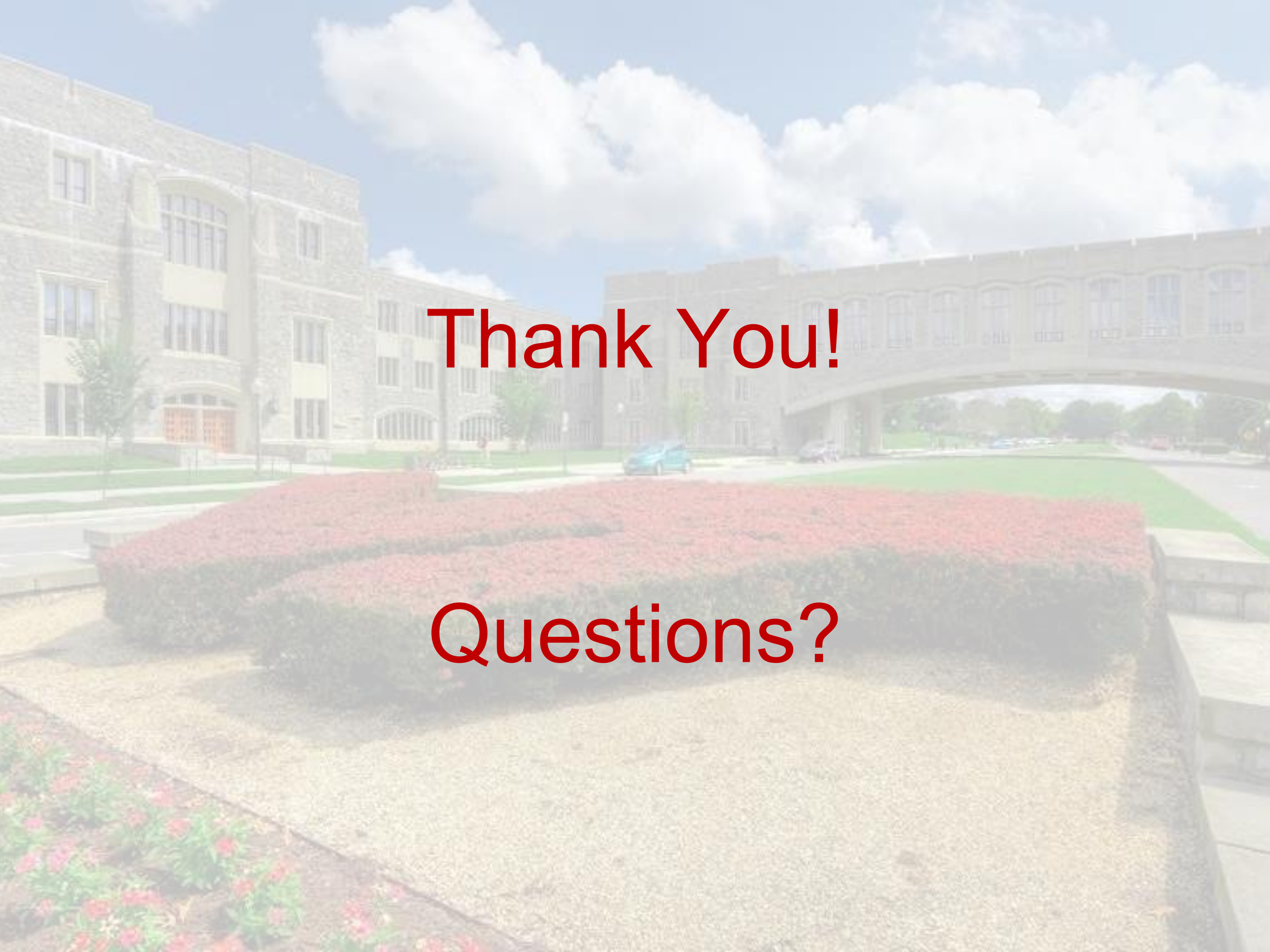


Starlink
Group 7-14

Falcon 9
Block 5
Feb 15, 2024.

Conclusion

- Starlink is a LEO satellite constellation providing broadband service across more global areas
- Unique components of Starlink include its orbital geometry, beam-steering technology, and ground infrastructure
- Combination of ground links and optical satellite links allow for low-latency service
- Future Developments center around providing more capacity



Thank You!

Questions?